Tutorial 1: Introduction

Exercise:

1. Given the problem: "For the given positive integer, justify if it is a prime."

   a) Formally define the problem

   Ans.

   Input: a positive integer n

   Output: Yes, if n is a prime; No, otherwise.

   (Or)

   Question: is n a prime?

   b) Give some instances and corresponding outputs

   Ans.

   Instance 1: a=10, output = No

   Instance 2: a=13, output = Yes

   c) Construct an algorithm and describe it with/without using pseudo code

   Ans.

   High-level description:

   Try to factor n by every positive integer a, for $2 \leq a \leq \sqrt{n}$.

   Pseudo code

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Algorithm 1: Prime(n)
Input: a positive integer n
Output: Yes, if n is a prime; No, otherwise.
Begin
   if n = 1 then
         return No
   end if
    for a = 2 to ⌊√n⌋ do
          if n%a = 0 then
                return No
          end if
    end for
    return Yes
end
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2. Suppose that $T_1(n) = O(f(n))$ and $T_2(n) = O(f(n))$. Which of the following are true? Justify your answer.

   a) $T_1(n) + T_2(n) = O(f(n))$

   Ans. True

   proof.

   Since $T_1(n) = O(f(n))$ and $T_2(n) = O(f(n))$, assume that

   1. $\forall n \geq n_1, T_1(n) \leq c_1 f(n)$ (1)
   2. $\forall n \geq n_2, T_2(n) \leq c_2 f(n)$ (2)

for some positive integer n_1, c_1, n_2, and c_2. Because n_1 and n_2 are only general notations, we assume $n_1 \geq n_2$ without loss of generality. Let $c = c_1 + c_2$. Then, for all $n \geq n_1$ we obtain

$$\begin{aligned} T_1(n) + T_2(n) &\leq c_1 f(n) + c_2 f(n) && \text{substituting (1) and (2)} \\ &\leq cf(n) && \text{definition of } c \end{aligned}$$

Thus, $\forall n \geq n_1, T_1(n) + T_2(n) \leq cf(n)$, where n_1 and c are two positive constants, which implies $T_1(n) + T_2(n) = O(f(n))$.

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b) $\frac{T_1(n)}{T_2(n)} = O(1)$

Ans. No

proof.

Let $T_1(n) = n^3$, $T_2(n) = n^2$, and $f(n) = n^3$. From the construction, we can see $T_1(n) = O(f(n))$ and $T_2(n) = O(f(n))$, but $\frac{T_1(n)}{T_2(n)} = n \neq O(1)$.

■

c) $T_1(n) = O(T_2(n))$

Ans. No, same as b).

3. Solve the following recurrence relations:

a) $T(1) = 1$, and for all $n \geq 2, T(n) = 3T(n-1) + 2$

Ans.

$$\begin{aligned} T(n) &= 3T(n-1) + 2 \\ &= 3(3T(n-2) + 2) + 2 \\ &= 3^2T(n-2) + 2 \times 3^1 + 2 \times 3^0 \\ &\;\;\vdots \end{aligned}$$

In general, after i iterations of substitutions

$$\begin{aligned} T(n) &= 3^iT(n-i) + 2(3^{i-1} + \cdots + 3^0) \\ &= 3^iT(n-i) + 2\frac{1-3^i}{1-3} && \text{geometric sequence} \end{aligned}$$

To eliminate i, let $i = n - 1$

$$\begin{aligned} T(n) &= 3^{n-1}T(1) + 2\frac{1-3^{n-1}}{1-3} \\ &= 3^{n-1} + 3^{n-1} - 1 \\ &= 2 \times 3^{n-1} \end{aligned}$$

b) $T(1) = 1$, and for all $n \geq 2, T(n) = 2T\left(\frac{n}{2}\right) + 6n - 1$

Ans.

$$T(n) = 2T\left(\frac{n}{2}\right) + 6n - 1$$

$$= 2\left(2T\left(\frac{n}{4}\right) + 6\left(\frac{n}{2}\right) - 1\right) + 6n - 1$$

$$= 2^2T\left(\frac{n}{2^2}\right) + 6n - 2 + 6n - 1$$

In general,

$$T(n) \quad = 2^iT\left(\frac{n}{2^i}\right) + i6n - (2^{i-1} + \cdots + 1)$$

Let $i = \log n$. Then, $2^i = 2^{\log n} = n$. By substitution,

$$T(n) \quad = nT\left(\frac{n}{n}\right) + 6n\log n - (n-1)$$

$$= n + 6n\log n - n + 1$$
$$= 6n\log n + 1$$

c) $T(1) = 1$, and for all $n \geq 2, T(n) = 3T\left(\frac{n}{2}\right) + n^2 - n$

Ans.

$$T(n) \quad = 3T\left(\frac{n}{2}\right) + n^2 - n$$

$$= 3\left(3T\left(\frac{n}{4}\right) + \left(\frac{n}{2}\right)^2 - \frac{n}{2}\right) + n^2 - n$$

$$= 3^2T\left(\frac{n}{2^2}\right) + \frac{3}{4}n^2 - \frac{3}{2}n + n^2 - n$$

$$= 3^2\left(3T\left(\frac{n}{2^3}\right) + \left(\frac{n}{2^2}\right)^2 - \frac{n}{2^2}\right) + \frac{3}{2^2}n^2 - \frac{3}{2}n + n^2 - n$$

$$= 3^3T\left(\frac{n}{2^3}\right) + \frac{3^2}{(2^2)^2}n^2 - \frac{3^2}{2^2}n + \frac{3}{2^2}n^2 - \frac{3}{2}n + n^2 - n$$

$$= 3^3T\left(\frac{n}{2^3}\right) + n^2\left(\frac{3^2}{(2^2)^2} + \frac{3^1}{(2^1)^2} + \frac{3^0}{(2^0)^2}\right) - n\left(\left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^1 + \left(\frac{3}{2}\right)^0\right)$$

$$= 3^3T\left(\frac{n}{2^3}\right) + n^2\left(\frac{3^2}{(2^2)^2} + \frac{3^1}{(2^2)^1} + \frac{3^0}{(2^2)^0}\right) - n\left(\left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^1 + \left(\frac{3}{2}\right)^0\right)$$

$$= 3^3T\left(\frac{n}{2^3}\right) + n^2\left(\left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^1 + \left(\frac{3}{4}\right)^0\right) - n\left(\left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^1 + \left(\frac{3}{2}\right)^0\right)$$

In general,

$$T(n) \quad = 3^iT\left(\frac{n}{2^i}\right) + n^2\left(\frac{1-\left(\frac{3}{4}\right)^i}{1-\frac{3}{4}}\right) - n\left(\frac{1-\left(\frac{3}{2}\right)^i}{1-\frac{3}{2}}\right)$$

Let $i = \log_2 n$. Then,

$$\frac{3^i}{4} \quad = \left(\frac{3}{4}\right)^i$$

$$= \left(\frac{3}{4}\right)^{\log_2 n}$$

$$= \left(\frac{3}{4}\right)^{\frac{\log_{\frac{3}{4}} n}{\log_{\frac{3}{4}} 2}}$$

$= (n)^{\log_2 \frac{3}{4}}$ (1)

Similarly, $\left(\frac{3}{2}\right)^i = (n)^{\log_2 \frac{3}{2}}$ (2)

And, $3^i = n^{\log_2 3}$ (3)

By substituting (1), (2) and (3),

$$T(n) = 3^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + n^2\left(\frac{1-n^{\log_2 \frac{3}{4}}}{1-\frac{3}{4}}\right) - n\left(\frac{1-n^{\log_2 \frac{3}{2}}}{1-\frac{3}{2}}\right)$$

$$= n^{\log_2 3} T(1) + 4n^2 - 4n^{2+\log_2 \frac{3}{4}} + 2n - 2n^{1+\log_2 \frac{3}{2}}$$

$$= n^{\log_2 3} + 4n^2 - 4n^{2+\log_2 3 - \log_2 4} + 2n - 2n^{1+\log_2 3 - \log_2 2}$$

$$= n^{\log_2 3} + 4n^2 - 4n^{\log_2 3} + 2n - 2n^{\log_2 3}$$

$$= 4n^2 - 5n^{\log_2 3} + 2n$$